

1. Physical Quantities.
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Prefix	Abbreviation	Power
kilo	K	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}
Peta	P	10^{15}
exa	E	10^{18}
zetta	Z	10^{21}
Yotta	Y	10^{24}
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
Pico	p	10^{-12}
femto	f	10^{-15}

1. Physical Quantity

- A physical quantity is any quantity that can be measured and expressed using a number and a unit. Eg. Length, mass, Time, Temperature, speed, Force, Energy, pressure et.c.
- # Without a unit a number has no physical meaning.

Types of physical quantities

A. Base quantities $\Rightarrow$ Base quantities are independent quantities that cannot be expressed in terms of other physical quantities. Eg. Length(m), Mass(kg), Time(s), Temperature(K), Electric current(A), Amount of substance(mol), Luminous intensity(cd).

B. Derived quantities $\Rightarrow$ Derived quantities are quantities that can be expressed in terms of base quantities. Eg. Area(m^2), volume(m^3), speed(m/s), Acceleration(m/s^2), Force(N or kgm/s^2), Energy(J or N.m), Power(W or J/s).

2. International system of units (SI Units).

Physical quantity	SI Unit	Symbol.
Length	meter	m
mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Amount of substance	mole	mol
Temperature	kelvin	K
Luminous Intensity	candela	cd

Standards of Length, mass and Time:

Standard of Length (meter): The distance traveled by light in vacuum during a time interval of $\frac{1}{299\,792\,458}$ second.

Standard of Time (second): The time taken 9192631770 oscillations of radiation emitted by cesium-133 atom.

Standard of mass (kg): The kilogram is the SI unit of mass is defined using fundamental physical constant:

(i) speed of Light (c)

(ii) Planck's constant (h)

(iii) cesium frequency ($\Delta\nu_{Cs}$)

Matter and model Building.

Matter: Anything that has mass and occupies space. (volume)

Eg. Books, solid, liquid, gases

Atomic Structure of matter:

(i) Protons: Positive charge.

(ii) Electrons: Negative charge

(iii) Neutrons: no charge

Atomic mass unit:

$$1u = 1.66 \times 10^{-27} \text{ kg.}$$

Models in physics:

A model is a simplified representation of physical system that helps us.

(i) Understand complex phenomena.

(ii) make predictions.

(iii) perform calculation easily.

☐ Density (ρ)

⇒ Density is defined as mass per unit volume of a substance

$$\rho = \frac{m}{V}$$

$m = \text{mass}$

$V = \text{volume}$

SI unit of density: kg/m^3

$$1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$$

If, two objects have same mass, the one with the lower density has larger volume.

$$\rho \propto \frac{1}{V}$$

Example:

Aluminium density = $2.7 \times 10^3 \text{ kg/m}^3$
 Iron density = $7.5 \times 10^3 \text{ kg/m}^3$

} Aluminium object has larger volume

Density of a cylinder

$$V = \pi r^2 h \quad \rightarrow \pi \text{ is a constant}$$

r is radius (m) } m.m } (m³)
 h is height (m) } m }

☐ Dimensional analysis:

The dimension of a physical quantity shows how it depends on base quantities.

Dimensions are written in square brackets $[]$

Fundamental dimensions:

Quantity

Symbol

Length

L

Mass

M

Time

T

Example: $v = at$.

L.H.S

$$v = \frac{s}{f} ; [v] = LT^{-1}$$

R.H.S

$$at \Rightarrow [a] = LT^{-2} ; [t] = T. \text{ so, } [at] = LT^{-2} \cdot T = LT^{-1}$$

so it is dimensionally correct.

Example: $v = at^2$

L.H.S

$$[v] = LT^{-1}$$

R.H.S

$$[a t^2] = LT^{-2} \cdot T^2 = L$$

so it is dimensionally incorrect.

Dimensionless Quantities:

- (i) angle.
- (ii) pure number
- (iii) Argument of trigonometric functions.

} Dimension 1.

Unit conversion:

Quantity

conversion

Quantity

conversion

1 km

1000 m

1 ft

0.3048 m

1 n

3600 s

1 mile

1.609 km

1 inch

2.54 cm

Speed Conversion:

km/h to m/s

5400 km/h to m/s.

$$= \frac{5400 \times 1000}{3600} \text{ m/s.}$$

$$= 1500 \text{ m/s.}$$

Practice: 1700 inch to meter.

$$1 \text{ inch} = 2.54 \text{ cm}$$

$$1700 \text{ inch} = (1700 \times 2.54) \text{ cm}$$

$$= 4318 \text{ cm}$$

$$= \frac{4318}{1000} \text{ m}$$

$$= 4.318 \text{ m.}$$

7000 ft to km.

$$1 \text{ ft} = 0.3048 \text{ m.}$$

$$7000 \text{ ft} = (0.3048 \times 7000) \text{ m}$$

$$= 2133.6 \text{ m}$$

$$= \frac{2133.6}{1000} \text{ km}$$

$$= 2.1336 \text{ km.}$$

Estimates and order of magnitude.

⇒ Used when an exact value is not required.

Estimate helping:

(i) Quick calculations.

(ii) checking if an answer is reasonable.

(iii) solving real life problems.

Rules to find order of magnitude

(1) Write the number in scientific notation.

(2) Compare the coefficient with 3.162.

if $a < 3.162$ order = 10^n

if $a > 3.162$ order = 10^{n+1}

Examples:

Number	scientific form	order of magnitude.
2700	2.7×10^3	10^3
2600	2.6×10^3	10^4
720	7.2×10^2	10^3
0.0086	8.6×10^{-3}	10^{-2}
0.0021	2.1×10^{-3}	10^{-3}

Practice problem from chapter:

$$a = k v^n \pi^m ; \text{ Find } (m, n)$$

L.H.S

$$[a] = L T^{-2}$$

R.H.S

$$[k v^n \pi^m] = 1 \cdot L^n T^{-n} \cdot L^m$$

$$= L^{n+m} \cdot T^{-n}$$

$$\text{So, } n = 2$$

$$\text{now, } m + n = 1$$

$$\Rightarrow m = 1 - 2 = -1$$

$$m = -1, n = 2$$

$$a = k \frac{v^2}{\pi}$$